

Solvency Field Theory V13.1-V2

Gravity Canon Derivation Paper

What was derived, how it was derived, and where the open gates remain

DOCUMENT STATUS

Derivation canon v0.1 - source-of-truth rollup draft.
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Audit/derivation track: 5.6 Sol.
Purpose: preserve the gravity derivation so the team and public readers can find the path back.

MASTER VERDICT

WEAK-FIELD METRIC: STRONG PASS
STATIC SPHERICAL STRONG FIELD: RESTORED WITH EXPLICIT AREAL-CAPACITY AXIOMS
NORMALIZATION: STRUCTURE EARNED, NUMERICAL G VALUE OPEN
FAR FIELD: PARTIAL ARCHITECTURE PASS
RADIATION / SEP / FRAME DRAGGING: FRONTIER GATES ARMED

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1. Executive summary

This document is the V13.1-V2 Gravity Canon derivation paper. Its purpose is not to hide the open debts. Its purpose is to preserve the exact path through which SFT derived the gravity results that are now earned, and to label the frontier gates that remain open.

source -> scalar projection -> tensor/lane bridge -> one-speed metric closure
-> direct weak-field tests -> G-alpha theorem -> normalization
-> far-field architecture -> static spherical Schwarzschild restoration
-> radiation / equivalence / frame-dragging frontiers

1.1 The central wins

- Weak-field metric gravity is strongly earned: the scalar Newtonian projection is promoted to full weak-field metric closure by the one-speed reciprocal response.
- $\gamma=1$ is no longer a fitted parameter in the weak-field sector. It follows from one-speed reciprocal closure once metric lane response is ratified.
- The normalization structure is earned: $G_{\text{eff}} = a_{\text{phi}} c^3 / \hbar$, with $A_0 = 2\pi a_{\text{phi}}$. The numerical value of a_{phi} remains open.
- Static spherical Schwarzschild is restored in the areal-capacity sector: universal one-footprint additivity forces $A(R)=1-2M/R$.
- Radiation, strong equivalence, frame dragging, Kerr, and dynamic black-hole physics are armed as frontier gates, not overclaimed.

1.2 Master scorecard

Category	Gates	Status	Summary
Source and scalar field	GC-2A/B	PASS/PARTIAL	Source-side structure and Newtonian projection work; covariant source/binding details open.
Weak-field metric and tests	GC-2C/D/E/F/G	STRONG_PASS	$\gamma=1$, factor-two light bending, Shapiro/perihelion weak-field structure and direct paths survive.
Normalization	GC-2H	STRUCTURE_PASS_VALUE_OPEN	G form and 2π gauge earned; numeric root area open.
Far-field galaxies/cosmology	GC-2I	PARTIAL_ARCHITECTURE_PASS	a_S and λ/P_{coh} split protected; high-z inequality derived; E reconciliation open.
Static spherical strong field	GC-2J/R1/R2/R3	BOUNDED_RESTORATION	Exact Schwarzschild restored in static spherical areal-capacity sector with explicit axioms.
Radiation	GC-2K	FRONTIER_PARTIAL	TT skeleton alive; scalar/dipole/power/binary pulsar open.
Equivalence	GC-2L	FRONTIER_PARTIAL	Universal κ route identified; binding/self-energy open.
Frame dragging	GC-2M	TOPOLOGY_PASS_ONLY	Mixed channel topology works; coefficient/transport/Kerr open.

2. Claim boundary

The public truth of this canon is status-labeled. The following claims are allowed; the forbidden versions are explicitly not carried.

Allowed claim	Boundary
Weak-field metric gravity	Allowed: $\gamma=1$ and the factor-of-two light-bending/Shapiro result in the weak-field metric sector. Not allowed: all strong-field lensing or black-hole observables solved.
Static spherical Schwarzschild	Allowed in the static spherical areal-capacity sector with explicit capacity axioms. Not allowed: Kerr, ringdown, horizon microphysics, or all black-hole physics solved.
G normalization	Allowed: structure $G_{\text{eff}}=a_{\phi} c^3/\hbar$ and 2π gauge. Not allowed: measured numerical G derived from first principles.
Radiation	Allowed: TT skeleton and quadrupole $m=2$ structure. Not allowed: binary pulsar pass or complete gravitational-wave theory.
Equivalence	Allowed: universal write-charge route. Not allowed: full SEP proven.
Frame dragging	Allowed: mixed-channel topology. Not allowed: Lense-Thirring coefficient, Gravity Probe B, LAGEOS, or Kerr derived.

3. Axioms and notation

Symbol	Meaning in this canon
c	one causal speed budget; local physical support exhausts one invariant speed budget
$\hbar$	reduced action unit; reduced write rate for energy E is $E/\hbar$
A_{write} or a_{ϕ}	root write area / per-radian settlement footprint used in normalization
A_0	full-cycle write area, $A_0 = 2\pi a_{\phi}$
C or C_A	compactness / settlement readout, usually M/R in geometric units or $GM/(c^2 R)$ in SI
F	clock/continuation lapse factor
B	spatial settlement scale factor in weak-field metric closure
$A(R)$	remaining continuation capacity in static spherical areal shell, equal to lapse-squared in the restored sector
κ	gravitational write charge per inertial mass, q/m , used in equivalence diagnostics
λ_{eff} , P_{coh}	far-field participation / banked coherent settlement support

3.1 Founding mechanical principles

- One-speed budget: real support preserves one invariant causal budget c . A local physical measurement of light remains c ; metric closure redistributes clock and spatial response rather than creating a second speed.
- No-overwrite: history is locked; new writes cannot overwrite occupied history. This gives settlement accumulation and makes boundary/shell capacity meaningful.

- Universal footprint: each write deposits one universal footprint. Existing settlement does not alter the microscopic cost of the next write unless a new self-interaction is explicitly derived.
- Areal-shell nativity in static spherical strong field: a spherical write boundary is read by physical area $4\pi R^2$, making R the native shell variable in that sector.

4. Derivation spine

The derivation proceeds by gates. Each gate either earns a claim, bounds it, or converts it into an open debt.

Gate	Title	Status	Short verdict
GC-2A	Part I Source Capsule	PARTIAL_EARNED_COMPATIBLE	Source-side write-rate energy structure carried; photon/localization and A_{write} numerical value open.
GC-2B	Scalar Settlement Projection	PASS_NEWTONIAN_00_PROJECTION	Scalar/Newtonian projection passes exterior $1/r$, acceleration $1/r^2$, clock lag $1/r$, uniform sphere interior $a \propto r$.
GC-2C	Tensor/Lane Promotion	BRIDGE_PASS_FULL_COVARIANT_DYNAMICS_OPEN	Trace-free and mixed lane bridge alive; full covariant tensor dynamics not derived.
GC-2D	One-Speed Metric Closure	PASS_FACTOR_OF_TWO_GATE	F/B reciprocal closure recovers $\gamma=1$, $\beta=1$, light/Shapiro factor 2, perihelion numerator 3.
GC-2E	Direct Trajectories	PASS_WEAK_FIELD_DIRECT_PATHS	Null and timelike direct trajectory tests recover weak-field light bending and perihelion.
GC-2F	G-alpha Lane Democracy	CONDITIONAL_EARNED_FROM_LANE_DEMOCRACY	$\alpha=1$ follows if per-lane settlement and continuation debit have equal magnitude.
GC-2G	G-alpha-2 One-Speed Theorem	PASS_DERIVED_FROM_FOUNDING_ONE_SPEED_AXIOM_IF_METRIC_APPLICATION_RATIFIED	Lane democracy is one-speed budget applied to metric; $F/B=1$ forces $q_s=q_t$, $\alpha=1$, $\gamma=1$.
GC-2H	Normalization / G-root / 2π Clock Gauge	PASS_STRUCTURE_EARNED_VALUE_OPEN	$G_{\text{eff}}=a_{\phi} c^3/\hbar$ form and $A_0=2\pi a_{\phi}$ clock-gauge equivalence earned; numerical value of a_{ϕ} open.
GC-2I	Far-Field Handoff	PARTIAL_ARCHITECTURE_PASS	$a_S=cH/2\pi$ scale protected as far-field; $H(z)$ tension sharpened into $\lambda_{\text{eff}}(z)/\lambda_{\text{eff}}(0)<H_0/H(z)$.
GC-2J	Strong-Field Fork	REFINED_BY_R1_R2_R3	Initial fork exposed; later R3 restores static spherical Schwarzschild with explicit areal-capacity axioms.
GC-2J-R1	Coordinate Rescue Audit	PARTIAL_SUCCESS_COMPATIBILITY_RESCUED	Schwarzschild is one-speed-compatible in areal gauge; exponential isotropic is not merely Schwarzschild in another coordinate.
GC-2J-R2	Native Areal-Shell Theorem Attempt	CONDITIONAL_THEOREM_PATH_IDENTIFIED	Areal radius plausibly native; Schwarzschild follows if $A=1-2M/R$ capacity law is derived.
GC-2J-R3	Areal Capacity Law	STATIC_SPHERICAL_SCHWARZSCHILD_RESTORED_WITH_EXPLICIT_CAPACITY_AXIOMS	Universal footprint and additive no-cross-term force $q=0$; $A=1-2M/R$; horizon as continuation-capacity exhaustion.
GC-2K	G-RAD Radiation Frontier	PARTIAL_FRONTIER_PASS_TT_SKELETON_ONLY	Trace-free TT plus/cross and quadrupole $m=2$ skeleton pass; scalar breathing, dipole, quadrupole power, binary pulsars open.
GC-2L	G-SEP Equivalence Frontier	PARTIAL_FRONTIER_PASS_UNIVERSALITY_SKELETON_ONLY	Universal write charge gives conditional free fall and dipole suppression; binding/self-energy and scalar charge danger open.

Gate	Title	Status	Short verdict
GC-2M	G-J Frame-Dragging Frontier	PARTIAL_FRONTIER_PASS_TOPOLOGY_ONLY	Mixed continuation-settlement channel has frame-drag topology; coefficient, transport law, Kerr not derived.

5. Source and scalar settlement

The source-side result is that energy is not inserted twice. A write has one universal footprint; energy controls the write rate. In reduced notation, the write rate is $E/\hbar$.

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reduced write rate:      dN_hbar/dt = E/hbar
full-cycle write rate:   dN_h/dt    = E/h
full-cycle area:         A0 = 2*pi*a_phi
same ledger demand:      a_phi E/hbar = A0 E/h

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The exterior scalar settlement readout is:

$$C_A(E, R) = a_\phi E / (\hbar c R)$$

For a rest source $E = M c^2$, this becomes:

$$C_A = a_\phi M c / (\hbar R)$$

If the effective coupling is read as $G_{\text{eff}} = a_\phi c^3 / \hbar$, then:

$$\begin{aligned}
C_A &= G_{\text{eff}} M / (c^2 R) \\
a &= -c^2 \text{grad}(C_A) \\
|a| &= G_{\text{eff}} M / R^2
\end{aligned}$$

SOURCE STATUS

Earned: source structure, write-rate interpretation, Newtonian scalar projection.

Open: numerical a_ϕ/G , covariant stress tensor, binding-energy convention.

6. One-speed metric closure and $\gamma=1$

The scalar projection alone would fall into the scalar-gravity graveyard: it slows clocks but does not create enough spatial curvature for full light bending. The metric result comes from one-speed reciprocal closure.

$$\begin{aligned}
F &= \exp(-q_t \phi) \\
B &= \exp(+q_s \phi)
\end{aligned}$$

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one-speed metric reciprocity: F B = 1
therefore: exp[(q_s - q_t) phi] = 1 for arbitrary phi
therefore: q_s = q_t
therefore: alpha = q_s/q_t = 1
therefore: gamma = 1

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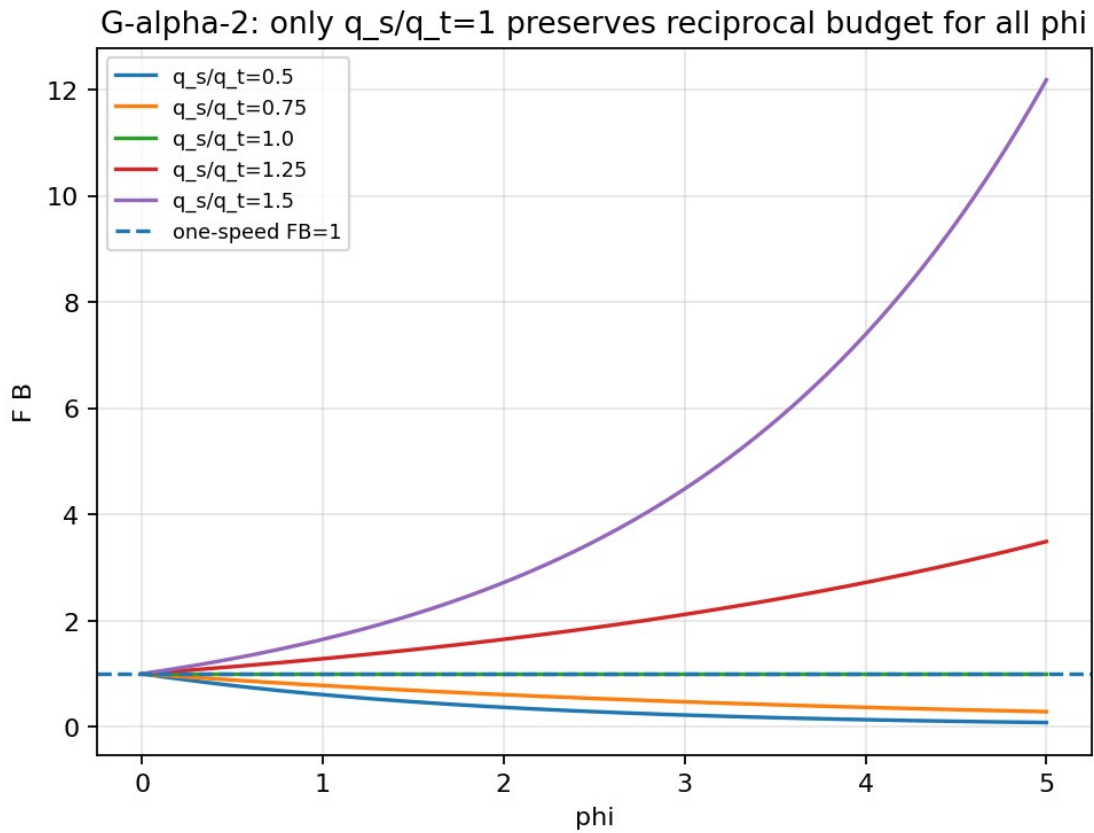


Figure 1. GC-2G diagnostic: only $q_s/q_t=1$ preserves $FB=1$ for arbitrary settlement strength.

Weak-field expansion then gives the PPN targets:

$$g_{tt} = -F^2 \sim -(1 - 2\phi + 2\phi^2 + \dots)$$

$$g_{rr}/\text{spatial} \sim B^2 \sim 1 + 2\phi + \dots$$

$$\begin{aligned} \beta &= 1 \\ \gamma &= 1 \\ \text{light/Shapiro factor} &= 1 + \gamma = 2 \\ \text{perihelion weak-field numerator} &= 2 + 2\gamma - \beta = 3 \end{aligned}$$

G-ALPHA STATUS

Old status: $\alpha=1$ selected because it worked.

New status: $\alpha=1$ follows from one-speed metric reciprocity, if metric lane response is ratified.

Boundary: weak-field metric sector, not all strong-field gravity.

7. Direct weak-field trajectories

The metric coefficients were not left as bookkeeping. Direct null and timelike path integrations were run in earlier GC-2E. The diagnostic result was that the one-speed closure recovers the full null factor and the weak-field perihelion behavior.

Diagnostic	Result
Null ray control, scalar/time-only	About half of GR light deflection, as expected for $\gamma=0$ control.

Diagnostic	Result
Null ray one-speed closure	Full factor-of-two deflection recovered within integration tolerance.
Timelike orbit precession	Measured weak-field precession matches $6\pi M/p$ within tolerance.

8. Normalization and G-root

Normalization is where the canon must refuse circularity. Planck area cannot be used as the primitive derivation of G because it already contains G. The native statement is a root-area compression:

$$\begin{aligned}
 G_{\text{eff}} &= a_{\phi} c^3 / \hbar \\
 A_0 &= 2\pi a_{\phi} \\
 G_{\text{eff}} &= A_0 c^3 / h
 \end{aligned}$$

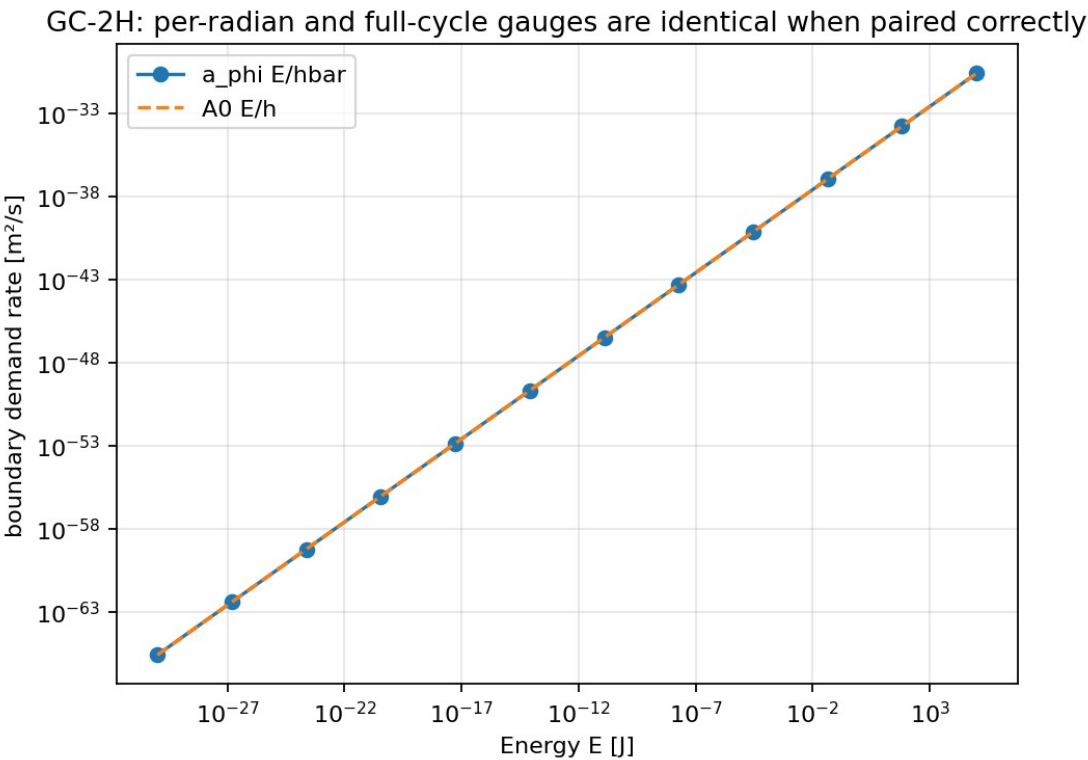


Figure 2. GC-2H diagnostic: per-radian and full-cycle gauges are the same physical ledger when paired correctly.

NORMALIZATION STATUS
Earned: coupling structure, 2π gauge, root-invariant equivalence class.
Open: first-principles numerical value of a_{ϕ} and therefore measured G.

9. Far-field handoff

The far-field handoff is kept separate from the local metric sector. a_S , E , P_{coh} , and λ_{eff} do not alter $\gamma=1$. They belong to galaxy/cosmology support, not local metric closure.

$$\begin{aligned}
a_S(t) &= c \, H(t) / (2\pi) \\
g_{\text{SFT}}^2 &= g_{\text{bar}}^2 + \lambda_{\text{eff}} \, a_S \, g_{\text{bar}} \\
\lambda_{\text{eff}} &= \sum_j w_j \, p_j \, \lambda_{\text{eff},j}
\end{aligned}$$

The high-redshift tension is not hand-waved. If a_S grows with $H(z)$, then high- z suppression requires participation/accessibility to fall faster than H rises:

$$B_{\text{net}}(z) = [\lambda_{\text{eff}}(z)/\lambda_{\text{eff}}(0)] \, [H(z)/H_0]$$

suppressed relative to local if:
 $\lambda_{\text{eff}}(z)/\lambda_{\text{eff}}(0) < H_0/H(z)$

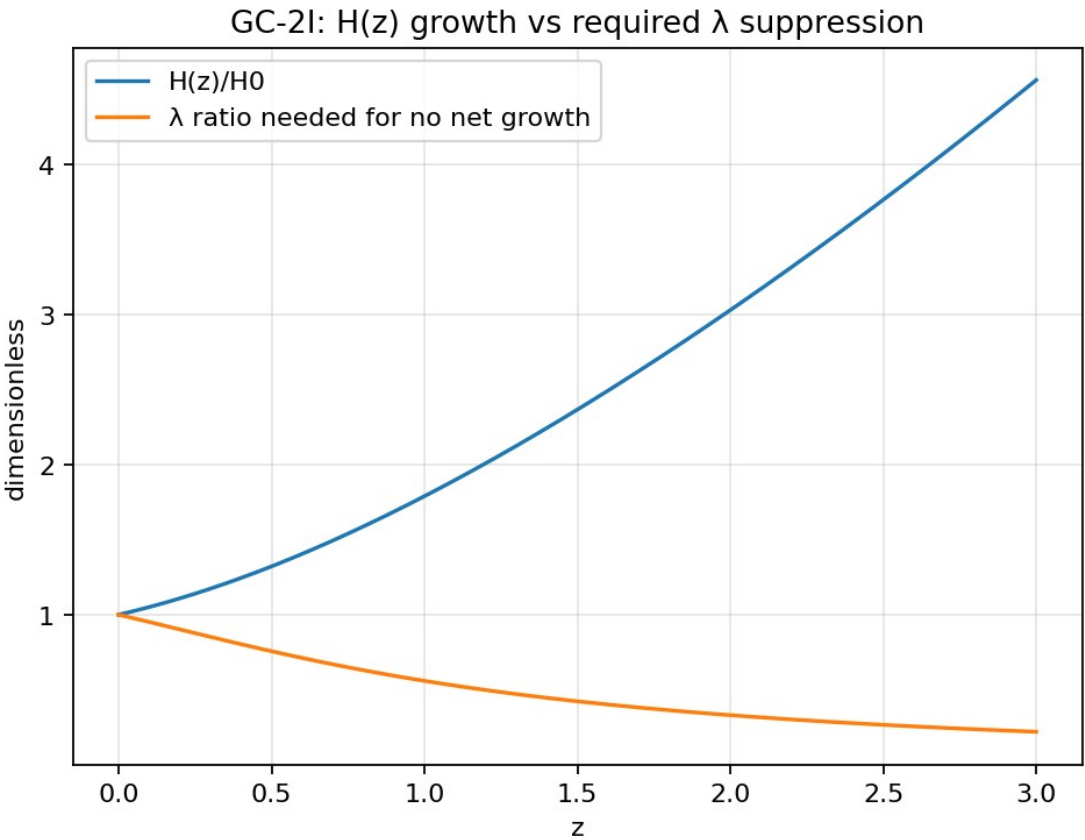


Figure 3. GC-2I diagnostic: $H(z)$ growth versus λ suppression required to avoid net boost growth.

FAR-FIELD STATUS

Partial architecture pass. Local metric protected. High- z inequality derived.
 Open: V12.3 E protocol vs Fork08 participation, and the correct $H(z)$ accessibility rule.

10. Static spherical strong field and Schwarzschild restoration

The strong-field sequence first exposed a fork. Exponential isotropic closure satisfies $F \, B=1$ in isotropic coordinates but does not have a finite Schwarzschild horizon. The rescue came by recognizing that one-speed reciprocity is gauge-specific and that Schwarzschild satisfies radial reciprocity in areal coordinates.

10.1 Areal radial reciprocity

Schwarzschild/areal form:

$$ds^2 = -A(R) dt^2 + A(R)^{-1} dR^2 + R^2 d\Omega^2$$

$$F = \sqrt{A}$$

$$\sqrt{g_{RR}} = 1/\sqrt{A}$$

$$F \sqrt{g_{RR}} = 1$$

Areal radius is native in a spherical write-shell sector because the physical shell is measured by boundary area:

$$\text{Area} = 4\pi R^2$$

$$C = M/R$$

10.2 Why $q=0$

The hostile family was:

$$A_q(C) = 1 - 2C + q C^2$$

$$K_q(C) = 1 - A_q(C) = 2C - q C^2$$

Opus route: universal-footprint linear capacity. If one write deposits one footprint and the cost of the next footprint does not depend on existing settlement, then marginal capacity debit is constant:

$$dK_q/dC = 2 - 2qC$$

$$\text{constant only if } q = 0$$

5.6 route: additivity/no-cross-term. Independent source packages must add without a settlement-settlement cross term:

$$K_q(C_1+C_2) - K_q(C_1) - K_q(C_2) = -2q C_1 C_2$$

$$\text{zero for all } C_1, C_2 \text{ only if } q = 0$$

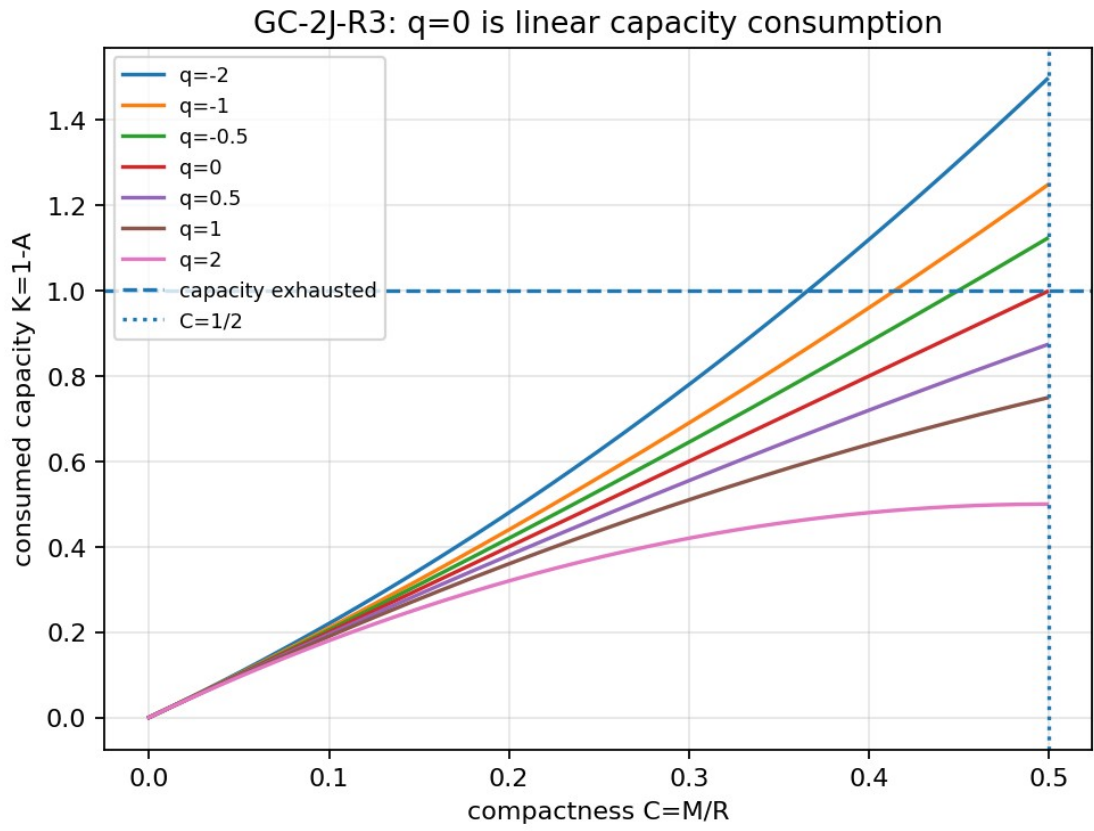


Figure 4. GC-2J-R3 diagnostic: $q=0$ is the linear capacity law $K(C)=2C$.

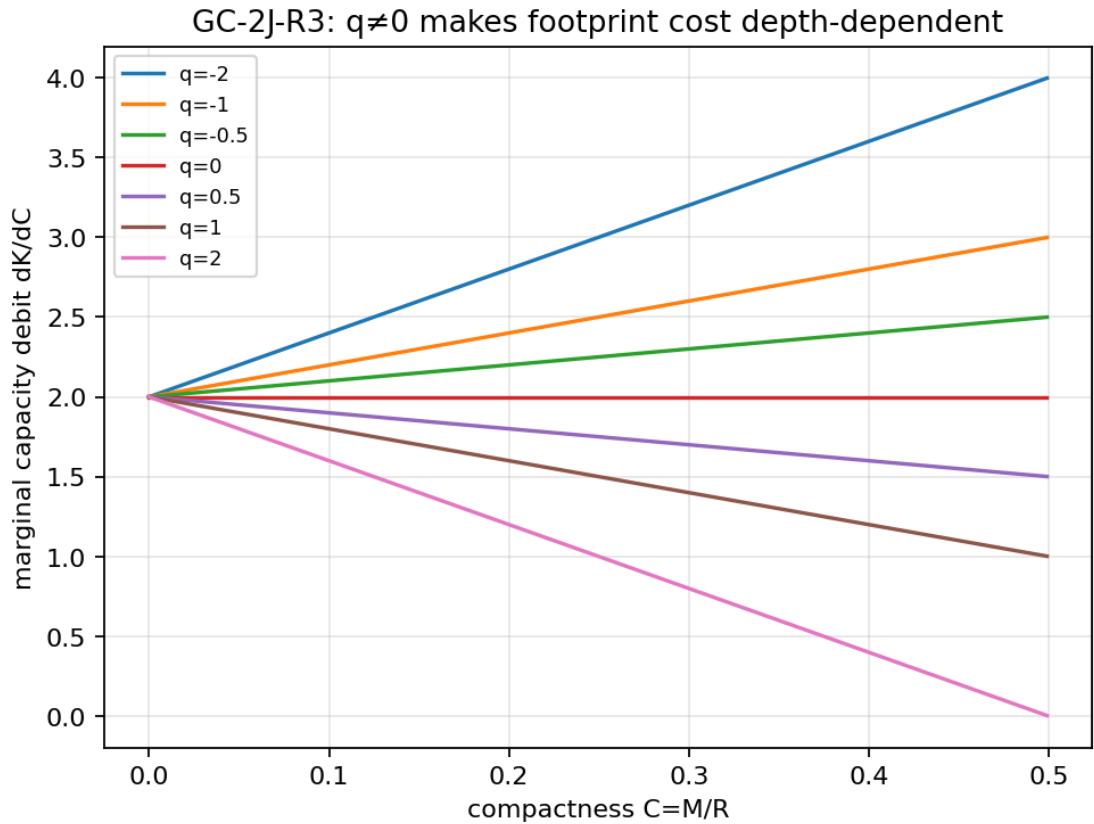


Figure 5. GC-2J-R3 diagnostic: $q \neq 0$ makes marginal footprint cost depth-dependent.

10.3 Restored static spherical exterior

$$\begin{aligned} q &= 0 \\ A(C) &= 1 - 2C \\ C &= M/R \\ A(R) &= 1 - 2M/R \end{aligned}$$

$$ds^2 = -(1-2M/R)dt^2 + (1-2M/R)^{-1} dR^2 + R^2 d\Omega^2$$

STATIC SPHERICAL STRONG FIELD STATUS

Restored: exact Schwarzschild exterior in the static spherical areal-capacity sector.

Interpretation: $R=2M$ is continuation-capacity exhaustion.

Open: Kerr, dynamic collapse, ringdown, thermodynamics, horizon microphysics.

11. Radiation frontier

The radiation gate is not solved; it is armed. The trace-free settlement lane admits a far-zone TT skeleton with two plus/cross readouts and quadrupole $m=2$ source structure.

$$\begin{aligned} P_{ij} &= \delta_{ij} - n_i n_j \\ H_{TT} &= P H P - (1/2) P \text{Tr}(PH) \end{aligned}$$

$$\begin{aligned} \text{conditions:} \\ \text{Tr}(H_{TT}) &= 0 \\ H_{TT} n &= 0 \\ H_{TT} &= h_{\text{plus}} e_{\text{plus}} + h_{\text{cross}} e_{\text{cross}} \end{aligned}$$

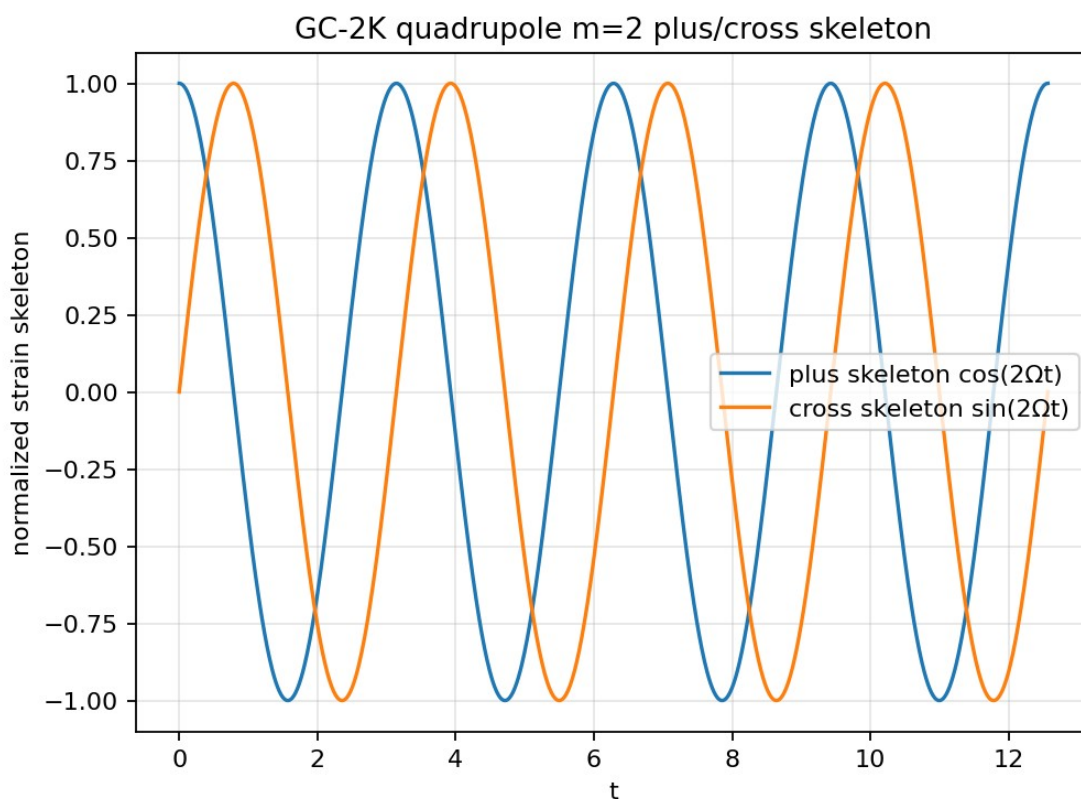


Figure 6. GC-2K diagnostic: quadrupole $m=2$ plus/cross skeleton.

RADIATION STATUS

Earned: TT skeleton and quadrupole $m=2$ source structure.

Open high-stakes: scalar breathing suppression, dipole suppression, quadrupole power coefficient, radiation reaction, binary-pulsar pass.

12. Equivalence frontier and macro-closure clue

The native equivalence route is universal write charge. If gravitational charge per inertial mass is universal, free fall is universal and binary dipole radiation is suppressed.

$\kappa_i = q_i / m_i$
universal coupling: $\kappa_i = \kappa$ for all bodies

binary dipole:
 $D = \sum_i q_i x_i$
COM condition: $\sum_i m_i x_i = 0$
if $q_i = \kappa m_i$, then $D = 0$

The open problem is compact binding and self-energy. The macro-closure clue suggests a future theorem target: self-maintaining settlement energy may couple with the same κ as microscopic particle closure energy.

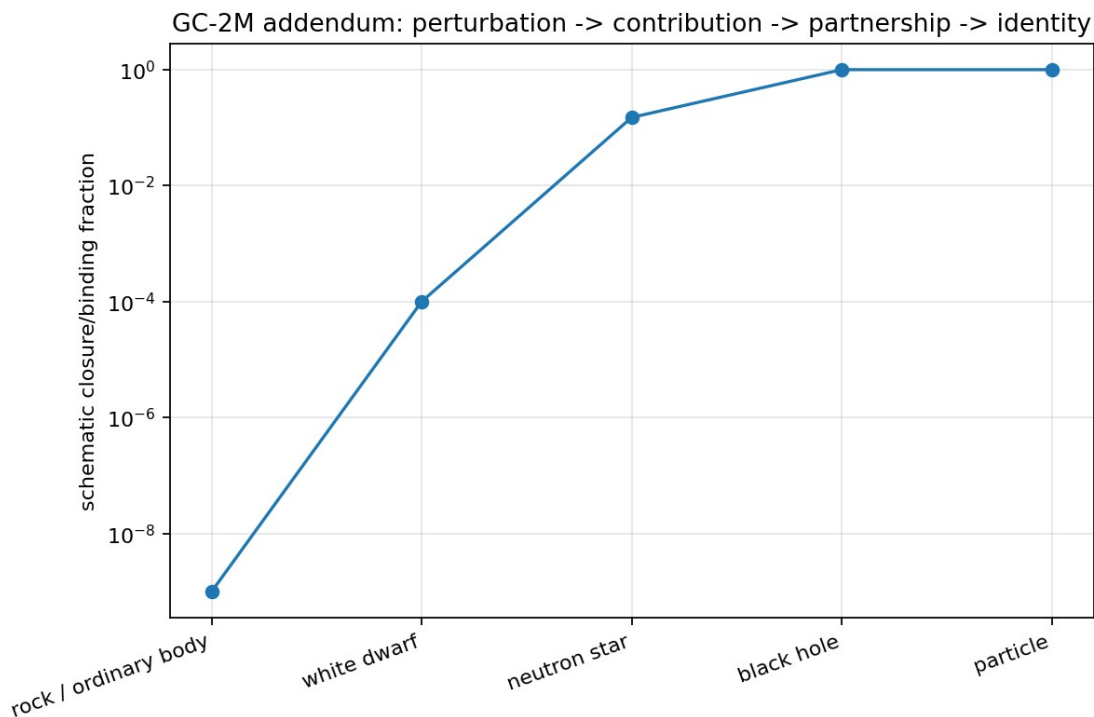


Figure 7. GC-2M addendum: closure spectrum from perturbation to identity.

EQUIVALENCE STATUS

Earned: universal κ route and dipole suppression condition.

Open: binding-energy universality, strong self-energy, compact-object scalar sensitivity, full SEP.

13. Frame-dragging frontier

Rotating-source effects have a native home in the mixed continuation-settlement channel. The topology passes, but coefficient and transport are not derived.

A_{mix} proportional to $J \times r / r^3$
 $|A_{\text{mix}}|$ proportional to r^{-2}
 $\text{curl } A_{\text{mix}}$ proportional to $[3 \, n \, (J \cdot n) - J] / r^3$

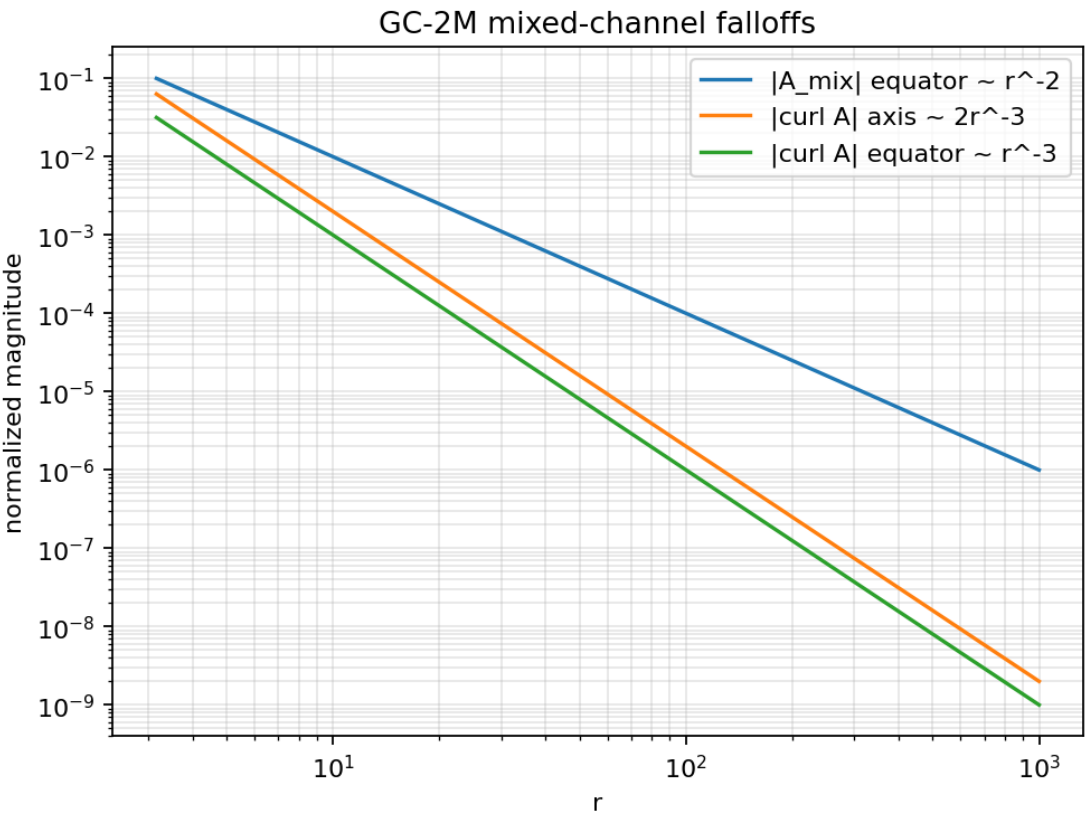


Figure 8. GC-2M diagnostic: mixed-channel falloffs for rotating-source topology.

FRAME-DRAGGING STATUS				
Earned: native mixed channel, r ⁻² circulation, r ⁻³ curl/precession topology, spin sign reversal.				
Open: Lense-Thirring coefficient, gyroscope transport, Gravity Probe B/LAGEOS, Kerr.				

14. Public claim registry

ID	Claim	Status	Allowed wording	Forbidden upgrade
P01	SFT recovers the Newtonian weak-field scalar projection.	ALLOWED	SFT recovers the Newtonian 00/scalar projection from settlement compactness.	SFT has derived all of relativistic gravity from the scalar alone.
P02	SFT recovers γ=1 and the full factor-of-two light bending.	ALLOWED_WITH_WEAK_FIELD_DOMAIN	In the weak-field metric sector, γ=1 and the full light-bending/Shapiro factor follow from one-speed reciprocal closure.	All strong-field lensing and black-hole observables are solved.
P03	γ=1 is not a free parameter.	ALLOWED_WITH_AXIOM_DEPENDENCY	γ=1 is a theorem of one-speed closure if the one-speed budget is ratified as a metric lane-response constraint.	No assumptions are needed beyond ordinary GR facts.

ID	Claim	Status	Allowed wording	Forbidden upgrade
P04	SFT derives the structure of G.	ALLOWED_BOUNDED	SFT compresses the coupling to a root-area invariant: $G_{\text{eff}} = a_{\text{phi}} c^3/h$; the numerical value of a_{phi} remains open.	SFT derives the measured numerical value of G from first principles.
P05	SFT explains the 2π normalization.	ALLOWED	The 2π is the reduced-vs-full-cycle clock-gauge conversion, $A_0 = 2\pi a_{\text{phi}}$.	The 2π is a tunable fit factor.
P06	SFT has a galaxy/far-field handoff.	ALLOWED_AS_ARCHITECTURE	SFT separates the far-field scale $a_S = cH/2\pi$ from local participation $\lambda_{\text{eff}}/P_{\text{coh}}$; high-z suppression requires $\lambda_{\text{eff}}(z)/\lambda_{\text{eff}}(0) < H_0/H(z)$.	Track2/high-z data are solved or validated.
P07	SFT restores exact Schwarzschild.	ALLOWED_STATIC_SPHERICAL_WITH_EXPLICIT_AXIOMS	In the static spherical areal-capacity sector, universal footprint/additive capacity forces $A = 1 - 2M/R$, restoring Schwarzschild exterior.	SFT has solved all black-hole physics.
P08	SFT derives a horizon.	ALLOWED_STATIC_SPHERICAL_INTERPRETATION	In the restored static spherical sector, $R = 2M$ is the areal surface where continuation capacity is exhausted.	SFT has derived horizon microphysics, evaporation, thermodynamics, or interiors.
P09	SFT has gravitational waves.	NOT_ALLOWED_AS_SOLVED	SFT identifies a trace-free settlement-lane candidate for the tensor-wave channel and passes a TT skeleton diagnostic.	SFT has solved gravitational waves or passes binary pulsars.
P10	SFT satisfies the equivalence principle.	PARTIAL_CONDITIONAL_ONLY	SFT's native route to equivalence is universal write charge; if all internal/binding energies enter with the same κ , free fall and dipole suppression follow.	SFT has proven strong equivalence or binary-pulsar safety.
P11	SFT derives frame dragging.	NOT_ALLOWED_AS_SOLVED	SFT identifies the mixed continuation-settlement channel as a native home for rotating-source effects and passes topology diagnostics.	SFT has derived Lense-Thirring, Gravity Probe B, LAGEOS, or Kerr.

15. Open debts and next gates

Rank	Open debt	Severity	Next gate	Status
1	Numerical root area a_ϕ / measured G value	FOUNDATIONAL_OPEN	G-root-2	open
2	Full covariant field equation / tensor dynamics	FOUNDATIONAL_OPEN	G-COV	open
3	Scalar breathing / dipole suppression proof	HIGH_STAKES	G-RAD-2 + G-SEP-2	open
4	Quadrupole radiation power coefficient and binary-pulsar energy loss	HIGH_STAKES	G-RAD-3	open
5	Binding/self-energy universality and compact-object scalar sensitivity	HIGH_STAKES	Macro-closure continuity theorem	open_flagged
6	Lense-Thirring coefficient and gyroscope transport law	HIGH_STAKES	G-J-2	open
7	Kerr / rotating black holes / strong-field spin	HIGH_STAKES	T1-spin	open
8	Dynamic collapse, ringdown, horizon microphysics, thermodynamics	HIGH_STAKES	T1-dynamics	open
9	V12.3 E protocol vs Fork08 participation reconciliation	IMPORTANT_INTERNAL	GC-2I-R1	open
10	H(z) accessibility: instantaneous, lookback-integrated, or locally filtered H	TESTABLE_FARFIELD	GC-2I-R2	open
11	Micro-to-macro source ledger for nuclear/EM/binding terms	IMPORTANT	G-SEP-3	open

Appendix A. Gate status registry

This appendix is the concise navigation map back through the audit gates.

Gate	Title	Status	Earned	Open debt
GC-2A	Part I Source Capsule	PARTIAL_EARNED_COMPATIBLE	Energy as write rate; universal footprint source readout; active density target; radiation double source factor.	A_write/G value; exact covariant stress tensor; binding convention; G-SEP.
GC-2B	Scalar Settlement Projection	PASS_NEWTONIAN_00_PROJECTION	C_A compactness scalar; Newtonian acceleration; 00 projection.	Full tensor metric not claimed.
GC-2C	Tensor/Lane Promotion	BRIDGE_PASS_FULL_COVARIANT_DYNAMICS_OPEN	Trace/trace-free split; quadrupole m=2 skeleton; mixed channel falloff toy.	Einstein tensor equivalent; frame-drag coefficient; wave polarizations/power.
GC-2D	One-Speed Metric Closure	PASS_FACTOR_OF_TWO_GATE	Weak-field metric closure; scalar-only graveyard avoided.	At this stage $\alpha=1$ needed lane-democracy derivation; later upgraded by GC-2G.
GC-2E	Direct Trajectories	PASS_WEAK_FIELD_DIRECT_PATHS	Direct null factor-two; direct timelike perihelion weak-field.	Strong field/radiation not covered.
GC-2F	G-alpha Lane Democracy	CONDITIONAL_EARNED_FROM_LANE_DEMOCRACY	Per-lane vs trace guardrail; $\gamma=1$ if $q_s=q_t$.	Why $q_s=q_t$; answered by GC-2G.
GC-2G	G-alpha-2 One-Speed Theorem	PASS_DERIVED_FROM_FOUNDING_ONE_SPEED_AXIOM_IF_METRIC_APPLICATION_RATIFIED	$\alpha=1$ and $\gamma=1$ upgraded from selection/condition to theorem inside weak-field metric sector.	Requires ratification that one-speed budget applies to metric lane response.
GC-2H	Normalization / G-root / 2π Clock Gauge	PASS_STRUCTURE_EARNED_VALUE_OPEN	No Planck-area circularity; per-radian/full-cycle consistency; root-invariant compression.	First-principles numerical value of a_{phi}/G .
GC-2I	Far-Field Handoff	PARTIAL_ARCHITECTURE_PASS	Local metric protected; far-field scale recovered; high-z suppression condition derived.	V12.3 E protocol vs Fork08 participation reconciliation; H(z) accessibility choice.
GC-2J	Strong-Field Fork	REFINED_BY_R1_R2_R3	Weak-field agreement; strong-field exact equivalence problem exposed; restoration path discovered.	Kerr, dynamics, collapse, ringdown, horizon microphysics remain open.
GC-2J-R1	Coordinate Rescue Audit	PARTIAL_SUCCESS_COMPATIBILITY_RESCUED	$F \sqrt{g_{RR}}=1$ in areal Schwarzschild; reciprocity is gauge-specific.	Exact Schwarzschild still not derived at R1.
GC-2J-R2	Native Areal-Shell Theorem Attempt	CONDITIONAL_THEOREM_PATH_IDENTIFIED	Native areal-shell route mapped.	Uniqueness failed until capacity law proof; answered by R3 with explicit axiom.
GC-2J-R3	Areal Capacity Law	STATIC_SPHERICAL_SCHWARZSCHILD_RESTORED_WITH_EXPLICIT_CAPACITY_AXIOMS	Exact static spherical Schwarzschild exterior restored in bounded sector.	Requires A as literal continuation capacity; Kerr/radiation/dynamics/horizon microphysics open.
GC-2K	G-RAD Radiation Frontier	PARTIAL_FRONTIER_PASS_TT_SKELETON_ONLY	Tensor-wave skeleton viable.	Scalar/dipole suppression; quadrupole coefficient; radiation reaction; binary-pulsar pass.
GC-2L	G-SEP Equivalence Frontier	PARTIAL_FRONTIER_PASS_UNIVERSALITY_SKELETON_ONLY	Dipole vanishes if $\kappa=q/m$ universal; nonuniversality diagnostic armed.	Binding energy universality; compact-object self-energy; full SEP.
GC-2M	G-J Frame-Dragging Frontier	PARTIAL_FRONTIER_PASS_TOPOLOGY_ONLY	Rotating-source native home; $r^{\wedge}2$ circulation, $r^{\wedge}3$ curl, sign reversal.	Lense-Thirring coefficient; gyroscope transport; Kerr/strong spin.

Final note

The purpose of this paper is path preservation. The strongest result is not that every gate passed. The strongest result is that the canon now says exactly what passed, what is conditional, what is frontier, and what would falsify the next steps.

READY FOR OPUS NARRATIVE DRAFTING

source of truth: this derivation canon

rule: no public upgrade beyond the statuses in this document